

# Physics in Ancient India: An Introspective study

Presented By,

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## Abstract

This poster explores the contributions of ancient India to the field of physics, delving into the rich history and groundbreaking advancements that have shaped modern understanding of physical phenomena. From the concept of atoms proposed in the Vaisheshika school of philosophy to various mathematical calculation in vedas ancient Indian scholars laid foundation principles in mechanics, optics, and acoustics. This introspective study highlights key texts like Vedas, Upanishads and various astronomical treatises, illustrating the profound understanding of natural laws and the universe. Through this poster, I aim to shed light on the sophisticated scientific legacy of ancient India, demonstrating its enduring impact on contemporary physics and inspiring future exploration.

## Introduction

- Ancient India, a cradle of civilization, has a storied past rich with scientific inquiry and innovation. Among the various field, physics stood out as a domain where Indian scholars made a significant contributions that predate many modern discoveries.
- The intellectual traditions of India, encapsulated in texts like the Vedas, Upanishads, and various treatises, reveal a profound understanding of physical principles that continue to resonate in contemporary science. This study not only honors the legacy of ancient Indian physicists but also inspire a renewed interest in the historical root of scientific thought.

## Methodology

*The study of physics in ancient India involve a multidisciplinary approach, combining historical analysis, textual interpretation, and comparative studies. Our method include the following steps:*

### Litreature Review:

- **Primary sources :** Analyzing original Sanskrit texts such as Vaisheshika Sutra by Kanada, Aryabhata, and Brahmas-phutasiddhanta by Brahmagupta.
- **Secondary source:** Reviewing scholarly interpretations, commentaries, and translations of these ancient texts by historians of science and ideologists. The data of secondary sources are collected from ancient book, epics articles, websites, and web-based journals.

## Results and Discussion

We discuss some geneses of today's physics in ancient India, which is not known to most of the people in the world. Physics as a part of natural science was developed initially in ancient India. Here we highlighted some of the contributions of ancient India to the field of physics.

### 1. Atomic Theory/ The concept of anu(atom) from Vaisikha sutra 4<sup>th</sup> century.

2500 years before Dalton's atomic theory the Indian Sage **Kanada** propounded that **parmanu**(atom) is an indestructible particle of matter. The atom is indivisible because it is a state at which no measurement can be assigned. They observed that atoms which are very minute objects are invisible to the human being's naked eyes which come into being and vanish in an instant.

- **Vaisheshikas** further claimed that atoms of the same substance combined with each other create **dyvanuka**(diatomic molecules) and **tryanuka**(triatomic molecules).

### 2. Speed of light in Vedas:

In the Vedas, Yojana is a unit of distance and Nimisha is a unit of time. It means that sunlight traverses 2202 Yojanas in half Nimisha. Distance travelled=2202 Yojanas

1 Yojana=9 miles, 110 Yards=21,144.705 miles (Approx.)

Time taken 1/2 Nimisha=0.114286 seconds (Approx.)

Speed of light=185,016.169 miles / seconds.

Modern Value=186,282.397 miles / seconds.

Modern science claims the speed of light to be approximately

186,282.397 miles / second.

### 3. Distance from earth to sun in Hanuman Chalisa:

its said: (Tulsidas)

**“Jug Shashtra yojan per Bhanu,**

**Leeyo tahi madur fal janu !!”**

**1Jug =12000 years**

**1 Sahastra =1000**

**1Yojan=8 Miles**

**Jug x shastra x yojan = per Bhanu**

**12000 x 1000 x 8 miles =96,000,000 x 1.6 km**

**=153,600,000 km**

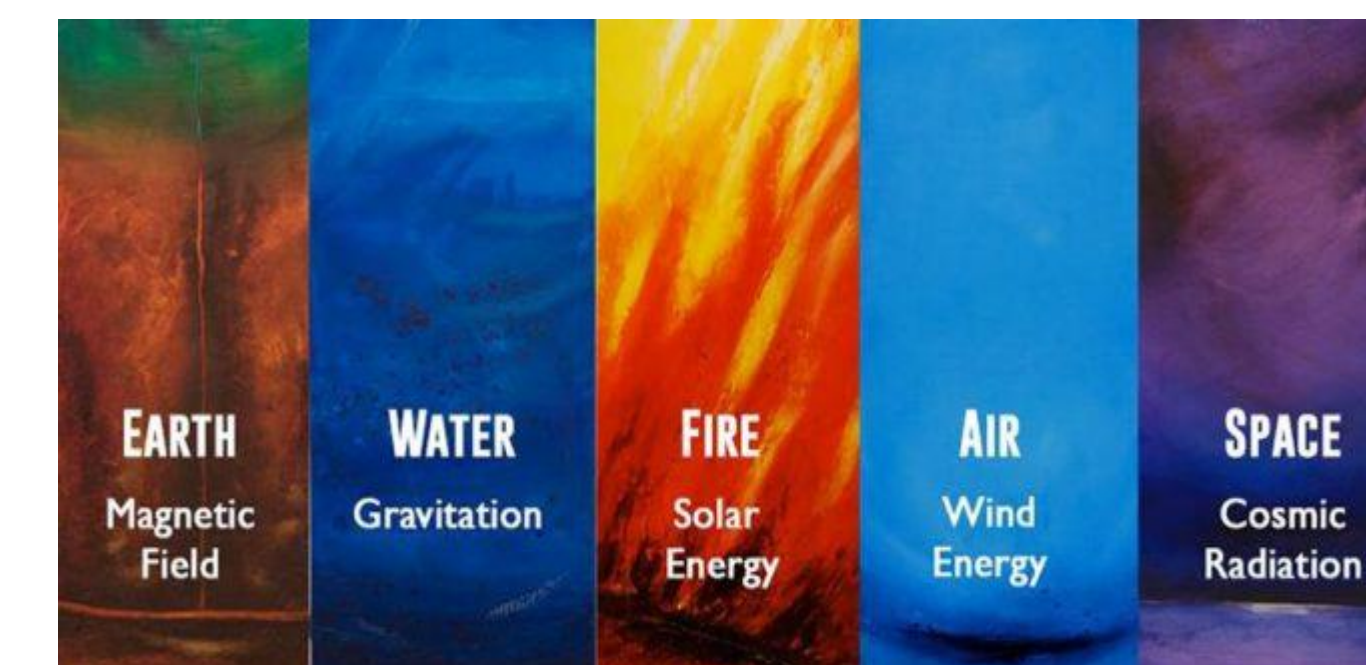
**According to NASA, it's almost 149,600,000 km.**

### 4. Theory of Panchabootha :

- This ancient Indian theory proposes that all of nature is composed of five elements: Akasha (Space), Prithvi (Earth), Vayu (Air), Agni (Fire), and Jala (Water).
- Akasha is considered unique, being eternal and the foundation for material objects.

### Comparison to Modern Science:

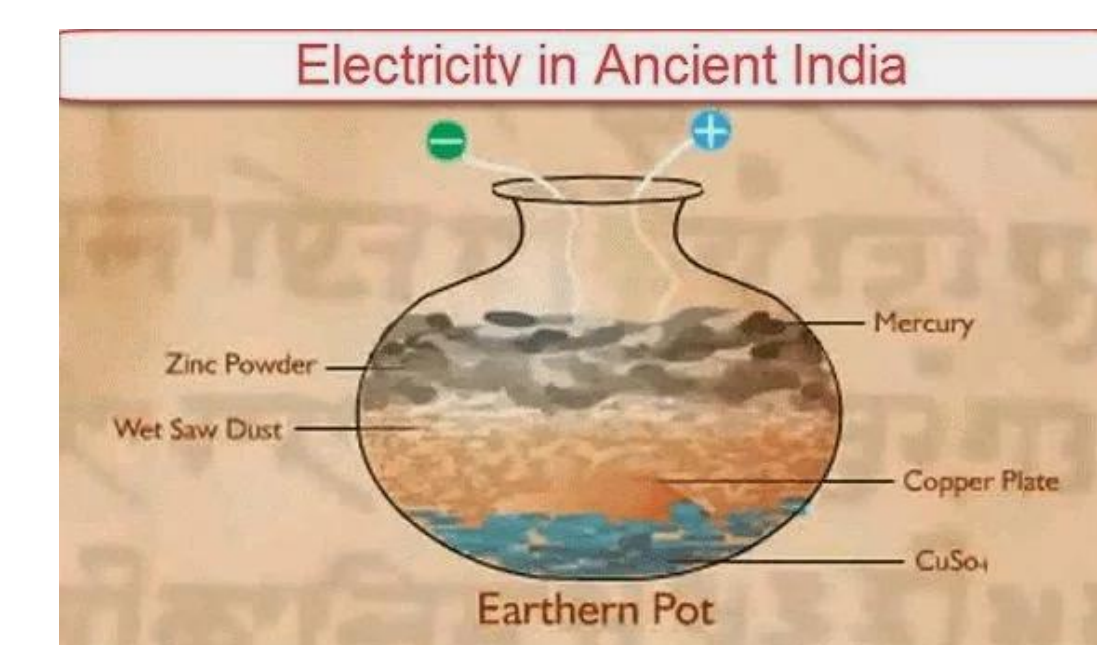
- The concept of Akasha is seen as similar to the historical idea of an all-pervading ether medium. (This concept has been largely abandoned in modern physics)
- The ancient text claims that the ideas of anu (molecules) and paramanu (atoms) predate Western atomic theory by Rishi Kanada.



**Fig 1.** The figure depicting different states of matter in ancient Indian text.

### 5. Electricity:

Agasthya samhita was an ancient book written by Rishi Agasthya around 7000 years ago. Agasthya explains the methodology involved in the construction of electric battery. He also described that water could be split into oxygen and hydrogen. An earthen pot was taken and covered with a clean copper plate. On the copper plate was the copper sulphate above which moist saw dust was placed. On top of all these, zinc amalgam sheet was placed known as Mitra–Varuna. Here Mitra means the cathode and varuna means anode. In order to intensify the power of electricity, a hundred of such jars known as Shata Kumbha were connected in series. When the reaction took place, water was split into Pranavayu (Oxygen) and Udanavayu (Hydrogen). The floating hydrogen was taken in air tight cloth and could be further used in aerodynamic applications.



**Fig 2.** The sketch of earthen pot producing electricity.

### 6. THEORY OF GRAVATION:

**Bhaskaracharya**, in his book Surya Siddhanta written in 11th century

the **Gravitational law** is states as

**“madhye samantandasya bhugolo vyomni tisthati bibhranah paramam saktim brahmano dharanatmikam”** [Surya Sidhantha 12th chapter 32 shloka].

**Translation:**The spherical earth stands at its centre in space due to the dharanatmikam sakti which prevents earth from falling away and helps it to stand firm.

**“akrsta saktisca mahi taya yat svastham guru svabhimukham svasaktya akrsyate tatpatativa bhati same samantat kva patatviyam khe”** [Sidhanta Shiromani, Bhuvanakosa, 6th shloka]

**Translation:** Every object falls on the ground due to earth's force of attraction. This force allows the sun, earth, moon and constellations to stay in the orbit.

**Shape of the earth:** Earth may appear flat but it is spherical in reality“.

He further explained this theory by stating: “if you draw a very big circle and look at one fourth of its circumference, you see it as a straight line.

But in true sense it is a circle. Similarly earth is spherical in shape.”

## Conclusion

The physics of ancient India, particularly duration the Vedic and post-Vedic period, contributed significantly to early scientific thought. Concept such cosmology, astronomy, mathematics, and metaphysics were explored in texts like Vedas, Upanishads, and later in work like Surya Siddhanta. Notable achievements include the calculation of planetary position. Kanada who gave the concept of atoms and atomic theory, also he gave the laws of motion. Bhaskaracharya who give the first principle of gravity in 11<sup>th</sup> century in his book Surya Siddhanta, the concept of wave particle dualism to the concept of sound phenomena was written in the texts like Vedas & Upanishad.

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